

Table D.4 Compressed Liquid (Water)*

Table D.4A Isobaric Data for $P = 5.0$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
273.15	5	1002.3	0.000998	0.044068	5.0	0.0001
280	5	1002.3	0.000998	28.731	33.7	0.1039
300	5	998.74	0.001001	112.15	117.2	0.3917
320	5	991.56	0.001009	195.5	200.5	0.66066
340	5	981.68	0.001019	278.9	284.0	0.91364
360	5	969.62	0.001031	362.5	367.7	1.1528
380	5	955.65	0.001046	446.5	451.7	1.38
400	5	939.91	0.001064	530.9	536.2	1.5968
420	5	922.46	0.001084	616.1	621.5	1.8047
440	5	903.27	0.001107	702.2	707.7	2.0053
460	5	882.22	0.001134	789.6	795.2	2.1998
480	5	859.08	0.001164	878.6	884.5	2.3897
500	5	833.51	0.001200	970.0	976.0	2.5764
520	5	804.92	0.001242	1064.3	1070.5	2.7618
537.09	5	777.37	0.001286	1148.2	1154.6	2.921

*Property values generated from NIST Database 23: REFPROP Version 7.0 (August 2002).

Table D.4B Isobaric Data for $P = 10.0$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
273.15	10	1004.8	0.0009952	0.1171	10.069	0.0003376
280	10	1004.7	0.0009954	28.659	38.613	0.10355
300	10	1001.0	0.0009991	111.74	121.73	0.39029
320	10	993.7	0.0010063	194.78	204.84	0.65846
340	10	983.84	0.0010164	277.92	288.08	0.91079
360	10	971.85	0.001029	361.27	371.56	1.1493
380	10	957.99	0.0010439	444.93	455.37	1.3759
400	10	942.42	0.0010611	529.06	539.67	1.5921
420	10	925.19	0.0010809	613.84	624.65	1.7994
440	10	906.28	0.0011034	699.52	710.55	1.9992
460	10	885.59	0.0011292	786.38	797.67	2.1928
480	10	862.94	0.0011588	874.8	886.39	2.3816
500	10	838.02	0.0011933	965.25	977.18	2.5669
520	10	810.36	0.001234	1058.4	1070.7	2.7504
540	10	779.15	0.0012835	1155.3	1168.1	2.9341
560	10	743.0	0.0013459	1257.6	1271.1	3.1213
580	10	699.05	0.0014305	1368.8	1383.1	3.3177
584.15	10	688.42	0.0014526	1393.5	1408.1	3.3606

Table D.4C Isobaric Data for $P = 15.0$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg · K)
273.15	15	1007.3	0.000993	0.17746	15.069	0.00044686
280	15	1007.0	0.000993	28.581	43.476	0.10316
300	15	1003.1	0.000997	111.34	126.29	0.38886
320	15	995.83	0.001004	194.1	209.16	0.65627
340	15	985.98	0.001014	276.99	292.2	0.90797
360	15	974.05	0.001027	360.07	375.47	1.1459
380	15	960.3	0.001041	443.45	459.07	1.3719
400	15	944.88	0.001058	527.26	543.13	1.5875
420	15	927.86	0.001078	611.68	627.85	1.7942
440	15	909.22	0.0011	696.94	713.44	1.9933
460	15	888.87	0.001125	783.3	800.18	2.186
480	15	866.68	0.001154	871.1	888.4	2.3738
500	15	842.36	0.001187	960.75	978.56	2.5578
520	15	815.52	0.001226	1052.8	1071.2	2.7395
540	15	785.52	0.001273	1148.2	1167.3	2.9208
560	15	751.28	0.001331	1248.3	1268.2	3.1042
580	15	710.78	0.001407	1355.3	1376.4	3.294
600	15	659.41	0.001517	1474.9	1497.7	3.4994
615.31	15	603.52	0.001657	1585.3	1610.2	3.6846

Table D.4D Isobaric Data for $P = 20.0$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg · K)
273.15	20	1009.7	0.00099	0.22569	20.033	0.00046962
280	20	1009.4	0.000991	28.496	48.31	0.10271
300	20	1005.3	0.000995	110.94	130.84	0.38741
320	20	997.93	0.001002	193.44	213.48	0.65408
340	20	988.09	0.001012	276.07	296.31	0.90516
360	20	976.22	0.001024	358.89	379.38	1.1426
380	20	962.58	0.001039	442.0	462.77	1.368
400	20	947.31	0.001056	525.5	546.62	1.583
420	20	930.48	0.001075	609.58	631.08	1.7891
440	20	912.09	0.001096	694.44	716.37	1.9874
460	20	892.08	0.001121	780.32	802.74	2.1794
480	20	870.3	0.001149	867.53	890.51	2.3662
500	20	846.53	0.001181	956.44	980.07	2.5489
520	20	820.44	0.001219	1047.6	1071.9	2.7291
540	20	791.5	0.001263	1141.6	1166.9	2.9082
560	20	758.86	0.001318	1239.7	1266.0	3.0885
580	20	721.04	0.001387	1343.5	1371.3	3.2731
600	20	675.11	0.001481	1456.8	1486.4	3.4682
620	20	613.23	0.001631	1588.6	1621.2	3.6891
638.9	20	490.19	0.00204	1786.4	1827.2	4.0156

Table D.4E Isobaric Data for $P = 30.0$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
273.15	30	1014.5	0.0009857	0.28791	29.858	0.0002688
280	30	1014.0	0.0009862	28.307	57.893	0.10164
300	30	1009.6	0.0009905	110.16	139.87	0.38444
320	30	1002.1	0.0009979	192.15	222.08	0.64972
340	30	992.24	0.0010078	274.29	304.52	0.89961
360	30	980.48	0.0010199	356.62	387.21	1.1359
380	30	967.04	0.0010341	439.19	470.21	1.3603
400	30	952.05	0.0010504	522.11	553.62	1.5742
420	30	935.59	0.0010688	605.53	637.6	1.7791
440	30	917.67	0.0010897	689.63	722.33	1.9761
460	30	898.25	0.0011133	774.62	808.02	2.1666
480	30	877.23	0.00114	860.75	894.95	2.3516
500	30	854.45	0.0011703	948.32	983.43	2.5322
520	30	829.66	0.0012053	1037.7	1073.9	2.7095
540	30	802.5	0.0012461	1129.5	1166.9	2.885
560	30	772.42	0.0012946	1224.4	1263.2	3.0601
580	30	738.56	0.001354	1323.4	1364.0	3.237
600	30	699.47	0.0014296	1428.5	1471.4	3.4189
620	30	652.41	0.0015328	1542.9	1588.9	3.6116
640	30	591.01	0.001692	1674.8	1725.5	3.8283

Table D.4F Isobaric Data for $P = 50.0$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
273.15	50	1023.8	0.000977	0.28922	49.126	-0.0010315
280	50	1022.9	0.000978	27.862	76.742	0.098824
300	50	1017.8	0.000982	108.63	157.76	0.37828
320	50	1010.1	0.00099	189.68	239.18	0.64102
340	50	1000.3	0.001	270.92	320.9	0.88875
360	50	988.72	0.001011	352.32	402.89	1.123
380	50	975.62	0.001025	433.9	485.15	1.3454
400	50	961.12	0.00104	515.76	567.78	1.5573
420	50	945.31	0.001058	597.99	650.88	1.7601
440	50	928.2	0.001077	680.73	734.6	1.9548
460	50	909.8	0.001099	764.15	819.1	2.1426
480	50	890.05	0.001124	848.42	904.59	2.3245
500	50	868.88	0.001151	933.75	991.29	2.5015
520	50	846.15	0.001182	1020.4	1079.5	2.6744
540	50	821.66	0.001217	1108.6	1169.5	2.8442
560	50	795.16	0.001258	1198.9	1261.8	3.012
580	50	766.3	0.001305	1291.7	1356.9	3.1789
600	50	734.55	0.001361	1387.6	1455.7	3.3463
620	50	699.21	0.00143	1487.7	1559.2	3.516
640	50	659.19	0.001517	1593.4	1669.3	3.6907